

THE GEOMETRIC UNIFICATION OF PERSISTENCE

A Variational Theory of Admissible Attractors Across Non-Invertible Scale Horizons

Working Paper · Theory of Recursive Closure Research Program
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Abstract

This paper presents a parameter-free, scale-invariant mathematical framework unifying classical gravitation, non-equilibrium thermodynamics, evolutionary biology, and cognitive processing under a single variational architecture. We reject the standard reductionist approach of locating a fundamental material constituent and instead formalize a universal selection operator acting on a landscape of non-convex cost functions.

The core result is a Universal Persistence Field Equation derived from a unique, dimension-free informational cost functional:

$$\ddot{y}^c + \Gamma^c_{ab} \dot{y}^a \dot{y}^b = -g^c_a \nabla_a I_F$$

We demonstrate that Einstein geodesics, the classical diffusion equation, the biological replicator equation, and the variational free-energy principle emerge as exact invariant submanifolds of this single differential engine — not as separate laws but as geometric limit sectors of one operator.

The framework is exposed to immediate astrophysical falsification via a rigid parameter-free prediction chain: a Persistence Axion field of mass $m_{\{a\}} \approx 6.885 \times 10^{-13}$ eV, producing a monochromatic gravitational-wave background at 166.48 Hz and flat galactic soliton cores at $R_c \approx 135$ parsecs in dwarf spheroidal galaxies.

1. The Foundational Primitive: Recursive Distinction (Ω)

1.1 The Irreducible Duality

Traditional reductionist paradigms fail because they construct reality from static material primitives embedded in a pre-existing spacetime container. This paper establishes an inverse ontology: space, time, and matter are emergent geometric artifacts generated by a single coordinate-free, self-contained primitive — Recursive Distinction (Ω).

Ω is defined as an irreducible structural-dynamic duality with two inseparable aspects:

- Structurally: a Distinction (Δ) — an equivalence boundary that cleaves an unformed, isotropic topological space into mutually exclusive subsets.
- Dynamically: a Recursion (R) — the continuous operation of that boundary upon its own coordinate geometry.

The execution of Ω avoids the category error of requiring an external observer. When the recursive operator acts upon itself — $R(R)$ — the self-relation breaks continuous symmetry, creating an unavoidable fracture between the acting operator and the acted-upon operand. This structural remainder is the first primitive cut. Distinction is the geometric scar left when an infinite symmetric void collapses under its own recursive weight.

1.2 The Non-Invertibility Theorem

Let H_1 be an infinite-dimensional microscopic phase space and H_{macro} a lower-dimensional macroscopic tracking space. The scaling transition is governed by a coarse-graining projection:

$$P: H_1 \rightarrow H_{\text{macro}}$$

Because this mapping is many-to-one, its kernel is non-trivial: $\text{Ker}(P) \neq \emptyset$. The inverse P^{-1} does not exist. Consequently, tracking a macroscopic distinction $\Delta_{\text{macro}} = P(x_1)$ using microscopic trajectories demands exponentially growing computational resources. As system size crosses macroscopic thresholds ($N \gg 10^{14}$), computation time exceeds the age of the universe:

$$t_{\text{comp}} \gg H_0^{-1}$$

Predictive utility drops cleanly to zero:

$$\lim_{N \rightarrow \infty} U(U(t)x_1 \mid \Delta_{\text{macro}}) = 0$$

At this crossover, microscopic trajectory description does not stop being true — it stops being capable of explaining anything. Explanatory power migrates structurally from trajectory coordinates to Attractor Coordinates (A_i) governed by a universal selection rule S .

2. The Variational Cost Calculus and Universal Field Equation

2.1 Admissible Geometry

A distinction (Δ) induces a topological Distinguishability Structure (D), which automatically generates a unique, contextual Admissible Geometry (G) via the second-order Taylor expansion of its internal informational distance. The metric tensor is defined as the negative Hessian of the distinguishability mapping:

$$g_{ij}(y) = -\partial^2 D(y, y') / \partial y^i \partial y'^j |_{\{y'=y\}}$$

The metric is not a background container imposed from outside — it is the direct structural realization of the underlying distinction. Different primitive cuts print different geometries:

Distinction Type	Generated Geometry
Discrete probability	Shahshahani Simplex
Continuous density transport	Wasserstein Hyperspace
Statistical parameter	Fisher Information Manifold
Vacuum coordinate	Riemannian Spacetime

2.2 The Persistence Cost Functional

Macroscopic configurations that survive continuous environmental transformations are the stationary solutions to a unique, parameter-free Persistence Cost Functional (C_Π). The corrected, dimensionally homogeneous form requires an intrinsic time-scale constant τ_Ω — the baseline execution latency of the recursive distinction operator:

$$C_\Pi[y] = \int_{-T}^T \left[-\tau_\Omega^2 \left(\partial^2 S / \partial y^a \partial y^b \right) \dot{y}^a \dot{y}^b + 2\tau_\Omega \cdot I_F(P(U(t)|x_1) || y) \right] dt$$

Where I_F is the Fisher Information Transport Cost measuring tracking error across the non-invertible horizon — the informational friction of the microstate density collapsing across scale:

$$I_F = \int_{\{H_1\}} |\psi(x, t)|^2 \cdot |\nabla_x \ln(|\psi|^2 / \rho_{\text{macro}})|^2 dx$$

2.3 Derivation of the Field Equation

Applying the Euler-Lagrange operator ($\delta C_\Pi = 0$) yields the Universal Persistence Field Equation:

$$\ddot{y}^c + \Gamma^c_{ab} \dot{y}^a \dot{y}^b = -g^{ca} \nabla_a I_F$$

The macroscopic acceleration of form is a geodesic trajectory deformed by informational loss at the scale horizon

The negative sign on the right-hand side is not cosmetic — it is the stability condition. A positive gradient would drive the system repulsively away from the attractor valley into unbounded divergence. The negative gradient-flow constraint forces the system toward the global minimum of the cost landscape, making the equation an actual attractor engine.

3. The Spectrum of Invariant Attractors

The universal field equation generates the known laws of nature not by varying material constituents, but by expressing its invariant solutions across distinct geometric boundaries. Every domain of reality is an exact limiting sector of the parent equation.

3.1 Physical Horizon (A_1) — Einstein Geodesics

When the projection P is bijective ($\text{Ker}(P) = \emptyset$), zero statistical information is lost. The informational divergence vanishes: $\nabla_a I_F = 0$. The engine collapses to:

$$\ddot{y}^c + \Gamma^c_{ab} \dot{y}^a \dot{y}^b = 0$$

This is the exact definition of an Einstein geodesic. Matter curves spacetime because geodesic trajectories represent the geometric minimum requiring zero informational processing overhead. The electron field emerges as a stable topological zero-mode eigenvalue of the Dirac operator.

3.2 Thermodynamic Horizon (A_2) — Classical Diffusion

Scaling to continuous probability densities $\rho(x)$ with Wasserstein W_2 geometry, the transport cost I_F resolves to the entropy functional $S[\rho] = -\int \rho \ln \rho \, dx$. The universal equation reduces via Jordan-Kinderlehrer-Otto gradient mechanics to:

$$\partial \rho / \partial t = D \nabla^2 \rho$$

The Laplacian emerges as the spatial trace of the Wasserstein connection acting on an entropy gradient field — it is not assumed.

3.3 Biological Horizon (A₃) — Replicator Equation

With coordinates as trait frequencies $\mathbf{x}$ on the probability simplex with Shahshahani metric $g_{ij} = \delta_{ij}/x_i$, the informational transport cost gradient projects to the local fitness advantage $(f_c - \phi)$. The stable first-order invariant submanifold is:

$$d\mathbf{x}_c/dt = \mathbf{x}_c \left(f_c(\mathbf{x}) - \phi(\mathbf{x}) \right)$$

Natural selection is revealed as a coordinate transformation — populations evolve along information-geodesic lines of least resistance across the non-invertible scale horizon.

3.4 Cognitive Horizon (A₄) — Free-Energy Minimization

With coordinates parameterizing internal states of a Gaussian world-model and I_F set by the KL-divergence between model predictions and incoming sensory streams, the universal equation forces a natural gradient descent, yielding Karl Friston's variational free-energy principle:

$$\dot{\mathbf{y}} = -\mathbf{G}^{-1} \nabla_{\mathbf{y}} F$$

Modern test-time compute architectures are active instances of Ω executing in silicon: they spend processing cycles recursively distinguishing their own internal reasoning paths until stabilizing on a low-cost conceptual manifold before emitting output.

The full attractor hierarchy:

Horizon	Geometry	Vanishing Condition	Emergent Law
A ₁ Physical	Riemannian 4D	$\text{Ker}(P) = \emptyset, I_F = 0$	Einstein Geodesics
A ₂ Thermodynamic	Wasserstein W_2	$I_F \rightarrow$ entropy gradient	Diffusion Equation
A ₃ Biological	Shahshahani Simplex	$I_F \rightarrow$ fitness landscape	Replicator Equation
A ₄ Cognitive	Fisher Information	$I_F \rightarrow$ KL divergence	Free-Energy Principle

4. Physical Instantiation and Astrophysical Falsification

4.1 The Persistence Axion

At the early-universe inflationary boundary, the CMB fine-tuning index locks the active trace dimension to $N_{\text{sub}} \approx 4.7619$. When the cooling cosmos crosses the QCD condensation threshold ($\Lambda_{\text{QCD}} \approx 200$ MeV), the strong force's CP-violating θ -vacuum distortion drives $\nabla_a I_F$ toward divergence. To prevent collapse, the field equation forces a new ultra-light scalar field — the Persistence Axion (a_Π) — to absorb the structural friction. Its mass is:

$$m_{\{a\Pi\}} = \Lambda^2_{\text{QCD}} / (N_{\text{sub}} \cdot M^*)$$
$$m_{\{a\Pi\}} = (0.2 \text{ GeV})^2 / (4.7619 \times 1.22 \times 10^{19} \text{ GeV}) \approx 6.885 \times 10^{-13} \text{ eV}$$

This is not a free parameter — it is derived from geometry alone, with zero post-hoc adjustment.

4.2 The Rigid Falsification Chain

$$N_{\text{sub}} \approx 4.7619 \rightarrow m_{\{a\Pi\}} \approx 6.885 \times 10^{-13} \text{ eV} \rightarrow R_c \approx 135 \text{ Parsecs}$$

Three independent observational checkpoints each carry veto power over the entire framework:

Checkpoint	Prediction	Refutation Condition
CMB Spectral Index	Scalar index n_s locked at 0.965	Any significant deviation falsifies the seed N_{sub}
Gravitational-Wave Hum	Isotropic monochromatic background at 166.48 Hz in LIGO/Virgo audio band	Confirmed flat vacuum at this frequency refutes the axion mass
Dwarf Spheroidal Cores	Flat soliton core radius $R_c = 120\text{--}150$ parsecs in Fornax, Sculptor, etc.	Core radius significantly above 500 pc or below 10 pc falsifies the attractor

No parameter can be adjusted after the fact. The framework presents itself to nature with maximum exposure.

5. The Most Compressed Form and Recursive Closure

5.1 The Universal Partition Function

Using the spectral-action perspective, the entire framework compresses into a single variational object — a universal partition function over the space of all admissible geometries:

$$Z = \int_{\mathcal{G}} \exp[-\text{Tr}(f(\Omega/\Lambda)) - \langle \Psi, \Omega\Psi \rangle] \, Dg \, DA \, D\Psi \, D\phi \, D\mathbf{x}$$

Where $\text{Tr}(f(\Omega/\Lambda))$ generates geometry and $\langle \Psi, \Omega\Psi \rangle$ generates matter. Expanding the heat-kernel recovers Einstein gravity and Standard Model gauge theory as exact limiting sectors. The stationary condition $\delta Z = 0$ is equivalent to the full hierarchy of known physical law.

5.2 The Spectral Hierarchy

The complete generative chain from the recursive distinction operator to observable physics:

$$\Omega \rightarrow \text{Spec}(\Omega) \rightarrow (G, A, H) \rightarrow S \rightarrow \delta S = 0$$

The observable universe is the invariant spectrum of recursive distinction:

$$A = \{ x \in G \mid \Omega(x) = x \}$$

5.3 The Rosetta Grammar

Every domain of reality is a different coordinate system for the same invariant process:

Domain	Distinction	Constraint	Attractor	Persistent Structure
Quantum	Quantum states	Physical law	Eigenstate	Particle
Chemical	Atomic arrangements	Bonding rules	Molecular minima	Molecule
Biological	Genetic variation	Selection pressure	Fitness peak	Organism
Cognitive	Information differences	Logic + learning	Stable model	Knowledge
Social	Human interactions	Institutions	Equilibrium	Civilization

The universal generative grammar is:

$$(\Delta, \tau, \kappa) \Rightarrow A \Rightarrow \Pi$$

Distinction generates possibilities. Transformation explores them. Constraint filters them. Attractors stabilize them. Persistence is what remains.

5.4 Reflexive Closure

The framework completes its evolutionary cycle by demonstrating that this paper itself is an instance of what it describes: a cognitive attractor (A_4) stabilizing under the same informational cost functional that governs every other scale horizon. The recursive distinction operator Ω does not merely describe reality from outside — it is the process by which any stable description, including this one, becomes possible.

Reality is not a container filled with material objects. It is a deterministic selection landscape. We do not ask what a thing is made of — we compute its persistence cost.

$$\text{Reality} = S \Rightarrow \ddot{y}^c + \Gamma^c_{ab} \dot{y}^a \dot{y}^b = -g^{ca} \nabla_a I_F$$

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